

Personalized Content Discovery *on the Netflix Prize*

"Turning 100 million ratings into the right next watch."

TEAM

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MEMBERS

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Discovery is two problems, not one

- **The job.** Learn $\hat{r}(u, i)$ from history, predict unseen ratings, then rank the catalogue into a personalised Top-10.
- **Two success axes that disagree.** *Rating accuracy* (RMSE) asks "is the predicted star value right?" — *Ranking quality* (MAP@10) asks "are the items we surface the ones they'll love?"
- **Our thesis.** Optimising one quietly sacrifices the other. We measure both, on the same models, and show the gap.

100M ratings, 98.7% empty

1.02_M

RATINGS USED

17.2_K

USERS

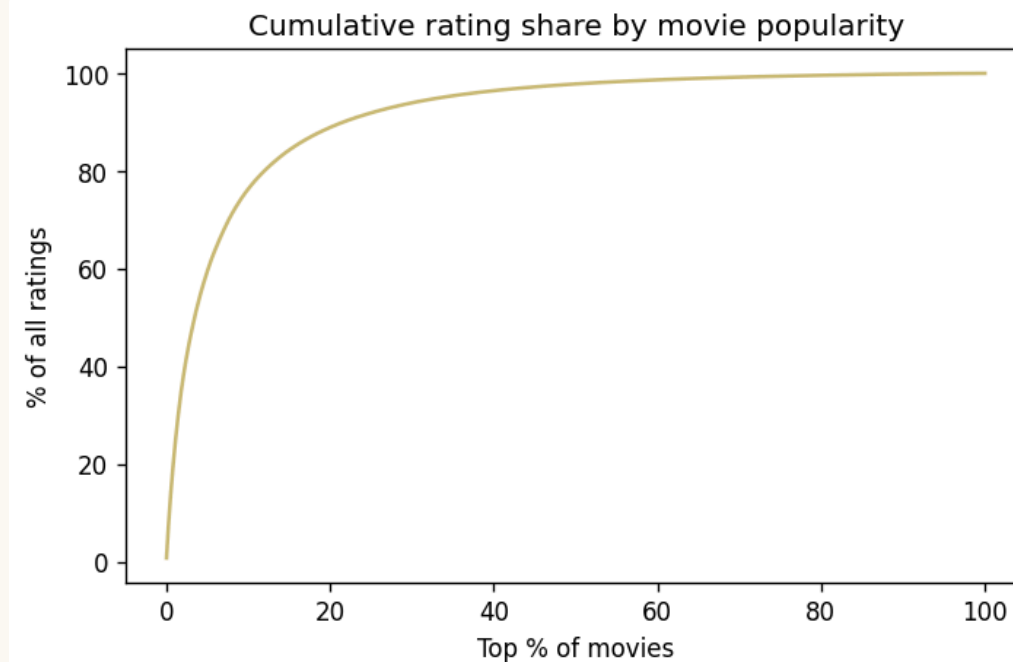
3.59

MEAN RATING

12%

MOVIES = 80% RATINGS

We sample 20K *users* (not rows) from combined_data_1 to preserve real per-user density — subsetting is explicitly allowed.



The long tail: 12% of movies absorb 80% of all ratings.

Four models, increasing sophistication

- **Baseline (bias)** — $\mu + b_u + b_i$, shrunk. Closed-form, fierce reference.
- **Item-Item CF** — adjusted cosine, top-50 neighbours, significance-shrunk.
- **Matrix Factorization** — 50-factor SGD with bias terms + validation early-stopping.
- **Ensemble** — non-negative stacked blend, weights learned out-of-fold. The Prize-winning move.

Per-user **temporal** split · relevance ≥ 3.5 · pure NumPy / scikit-learn · seed 42.

Accuracy and ranking crown different kings

MODEL	RMSE ↓	MAP@10 ↑	MAE ↓	NDCG@10	FIT
Baseline (bias)	0.9399	0.1843	0.7390	0.2899	0.1 s
Item-Item CF	0.9496	0.0854	0.7253	0.1532	18 s
Matrix Factorization	0.9118	0.1177	0.7110	0.2083	404 s
Ensemble (stacked)	0.9053	0.1419	0.7059	0.2384	74 s

HEADLINE

Best RMSE → the Ensemble (0.905). Best MAP@10 → the trivial Baseline (0.184). The most accurate model is *not* the best recommender.

Top-10, with reasons — and honest misses

User 1730413 — *"because you rated The Sum of All Fears & Braveheart highly"*

01	Lost: Season 1	3.88	05	CSI: Season 1	3.71
02	Firefly	3.81	06	My Favorite Wife	3.68
03	All Creatures Great & Small S3	3.75	07	The West Wing S3	3.67
04	Stargate SG-1 S7	3.74	08	Farscape: Peacekeeper Wars	3.67

✓ SUCCESS

Inu-Yasha (2000)

Predicted **4.26** · actual **5**. Surfaced a niche favourite.

✗ FAILURE

The Mummy (1999)

Predicted **4.60** · actual **2**. Popularity bias overrode taste.

When our ensemble learned to minimise *error*, it gave the baseline — our best *ranker* — a weight of **exactly zero**. Accuracy and discovery are different products.

- RMSE-best $\neq$ MAP-best, proven on our own models.
- Simplicity won ranking — at 1/4000th of MF's train cost.
- Caveat, disclosed: random negatives reward popularity.
- Tune on the goal — for discovery, that's ranking.

What we'd ship — and build next

- **Ship now.** Serve ranking from the bias baseline; serve rating estimates from the stacked ensemble. Both are cheap at inference.
- **Build next.** Rank-aware losses (BPR / WARP), ALS at scale, a metadata **hybrid** for **cold-start**, and coverage / diversity objectives for true discovery.
- **Reproducible by design.** One seeded command regenerates every number; full code at github.com/silversoul2213/ai-ml-cult.